Week 1 (task2)
Prompt Library v1

Reusable Prompt Templates for AI & Machine Learning Learning

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Introduction

Prompt engineering is the process of designing clear and effective prompts that help AI models generate accurate and useful responses. Instead of writing a new prompt every time, prompt engineers create reusable templates with placeholders that can be customized for different tasks.

This Prompt Library demonstrates a reusable prompt template for learning Artificial Intelligence and Machine Learning concepts. The same template is reused for different datasets and algorithms by changing only the variable values.

Reusable Prompt Template

Role

You are an expert AI and Machine Learning instructor with experience teaching university students.

Context

- Dataset: **{Dataset Name}**
- Algorithm: **{Algorithm Name}**
- Student Level: **Beginner**

Task

Explain the selected machine learning algorithm using the given dataset. Help the student understand both the theoretical concept and practical implementation.

Format

Generate the response using the following sections:

1. Introduction
2. How the Algorithm Works
3. Why this Algorithm is suitable for this dataset
4. Advantages
5. Limitations
6. Python Example using Scikit-learn

7. Real-world Applications
8. Interview Questions
9. Summary

Constraints

- Use simple English.
- Maximum 600 words.
- Use bullet points where appropriate.
- Include beginner-friendly explanations.
- Avoid unnecessary mathematical formulas.

Prompt 1

Variables

Dataset: Iris Dataset

Algorithm: Decision Tree

Prompt

Role:

You are an expert AI and Machine Learning instructor with experience teaching university students.

Context:

Dataset: Iris Dataset

Algorithm: Decision Tree

Student Level: Beginner

Task:

Explain the selected machine learning algorithm using the given dataset.

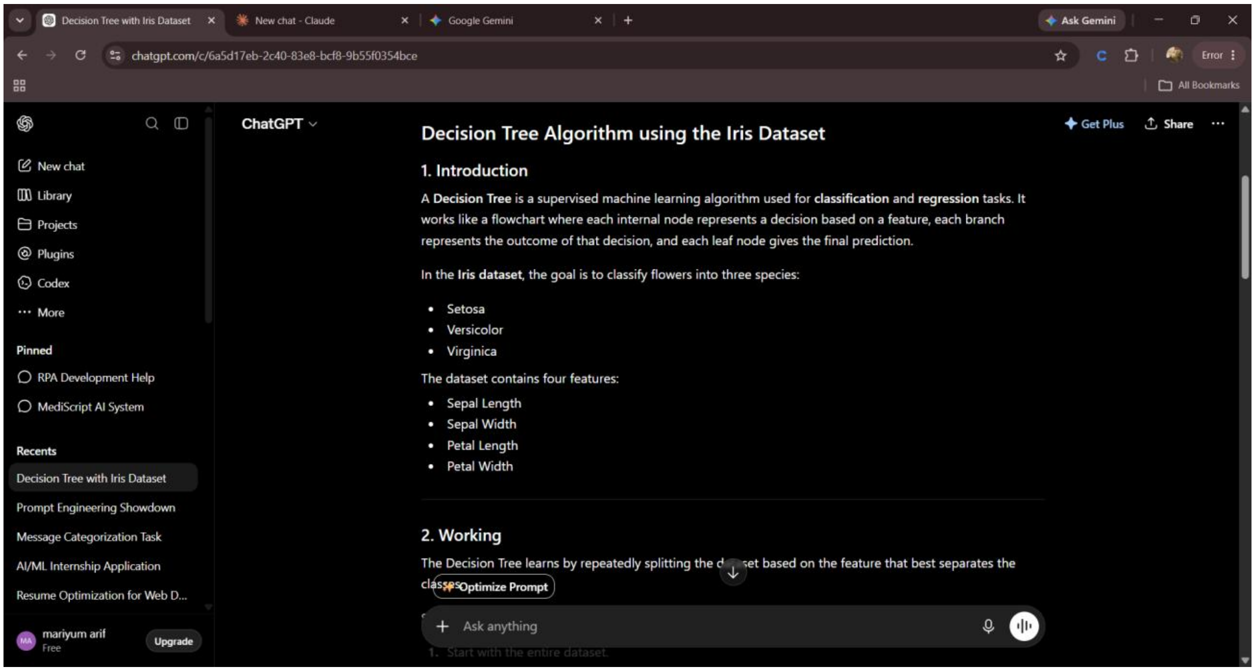
Format:

1. Introduction
2. Working
3. Why this algorithm fits this dataset
4. Advantages
5. Limitations
6. Python Example
7. Applications
8. Interview Questions
9. Summary

Constraints:

Simple English.

Maximum 600 words.



Decision Tree Algorithm using the Iris Dataset

1. Introduction

A **Decision Tree** is a supervised machine learning algorithm used for **classification** and **regression** tasks. It works like a flowchart where each internal node represents a decision based on a feature, each branch represents the outcome of that decision, and each leaf node gives the final prediction.

In the **Iris dataset**, the goal is to classify flowers into three species:

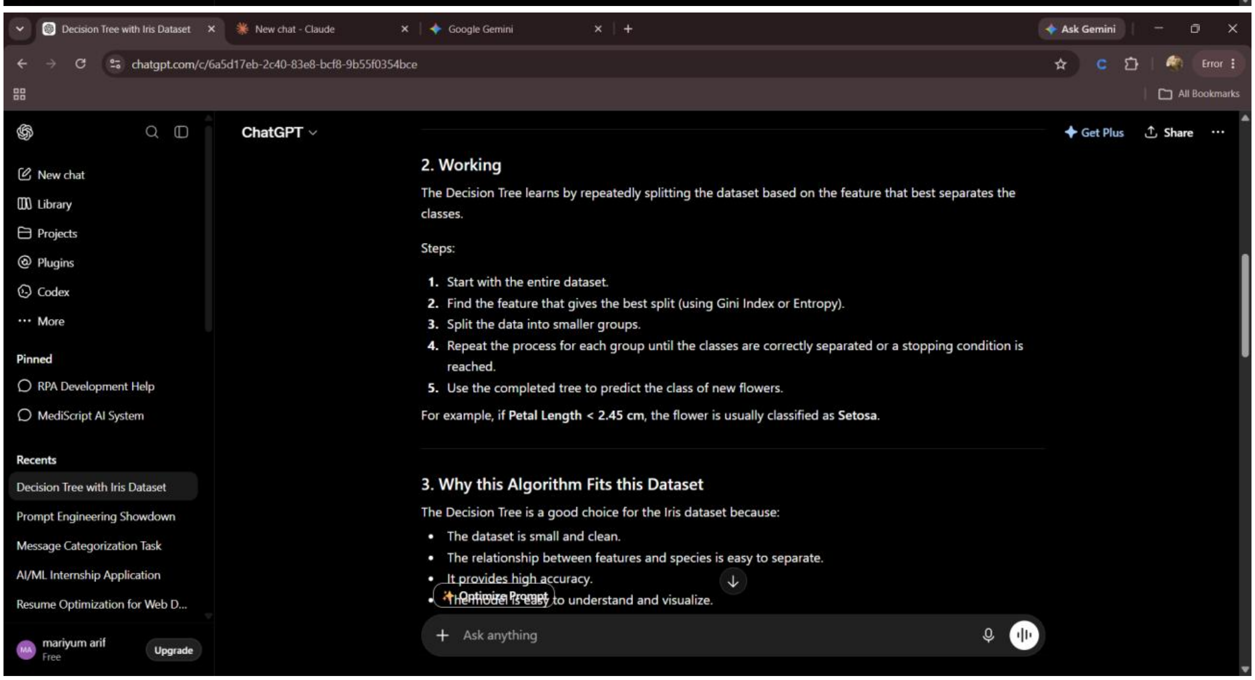
- Setosa
- Versicolor
- Virginica

The dataset contains four features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

2. Working

The Decision Tree learns by repeatedly splitting the d…set based on the feature that best separates the classes.



2. Working

The Decision Tree learns by repeatedly splitting the dataset based on the feature that best separates the classes.

Steps:

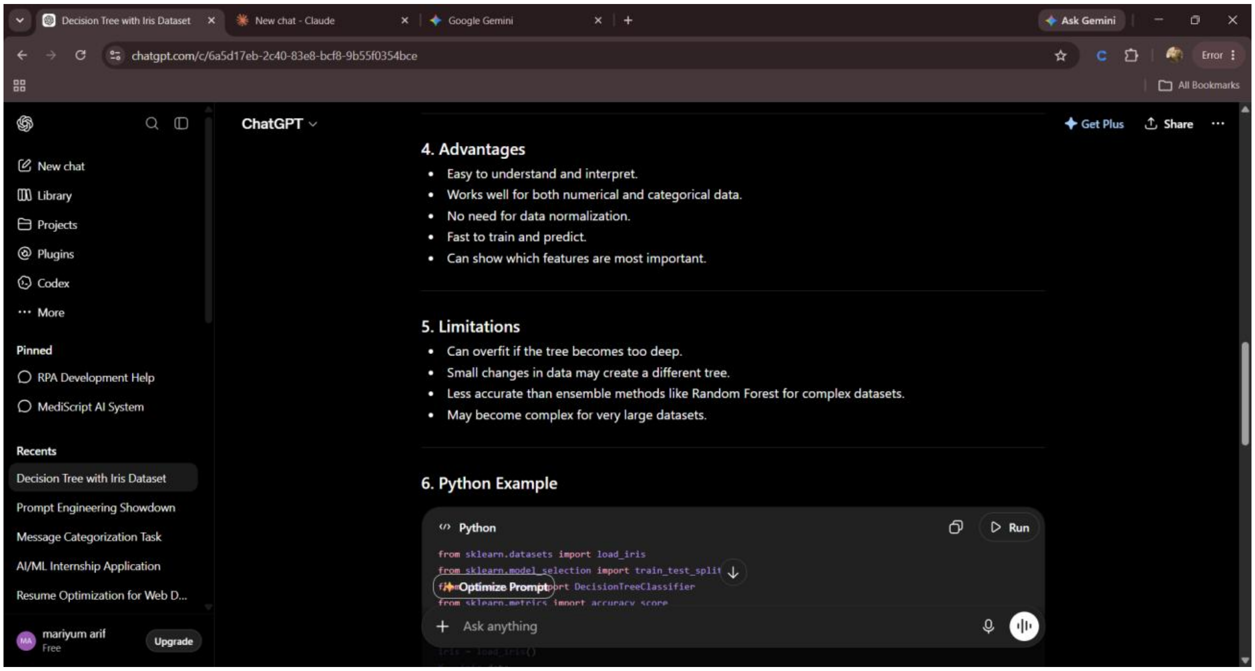
1. Start with the entire dataset.
2. Find the feature that gives the best split (using Gini Index or Entropy).
3. Split the data into smaller groups.
4. Repeat the process for each group until the classes are correctly separated or a stopping condition is reached.
5. Use the completed tree to predict the class of new flowers.

For example, if **Petal Length < 2.45 cm**, the flower is usually classified as **Setosa**.

3. Why this Algorithm Fits this Dataset

The Decision Tree is a good choice for the Iris dataset because:

- The dataset is small and clean.
- The relationship between features and species is easy to separate.
- It provides high accuracy.
- The model is easy to understand and visualize.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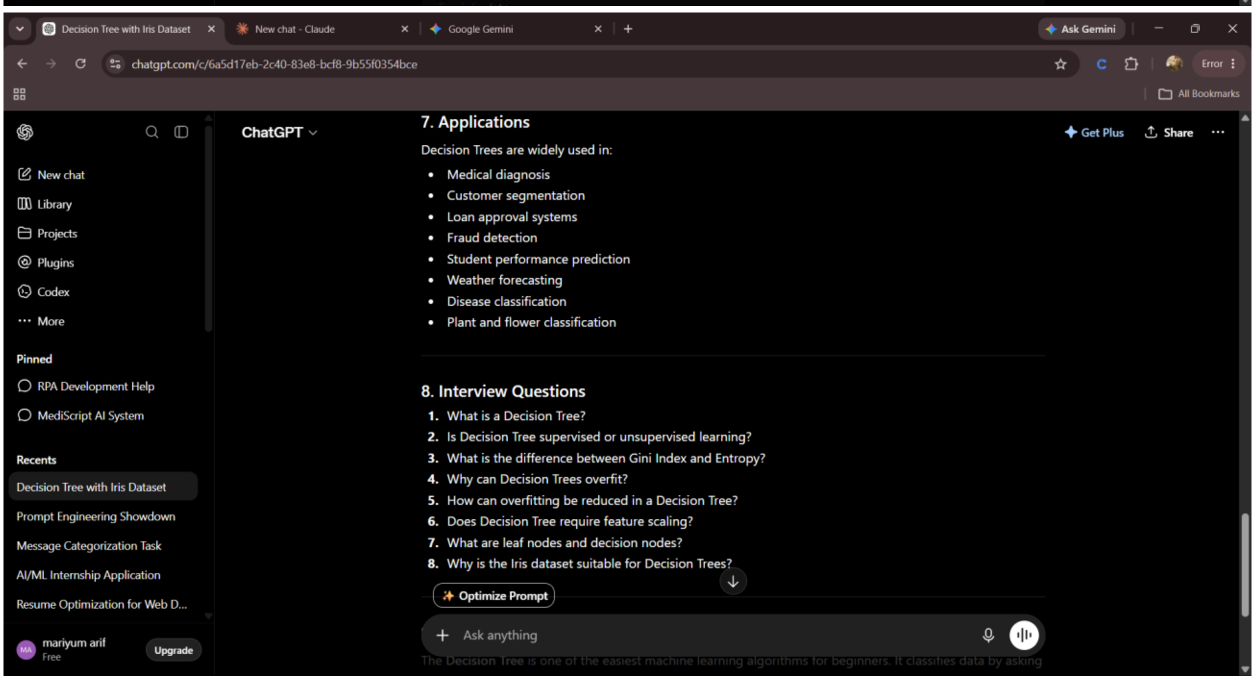
4. Advantages
Easy to understand and interpret.
Works well for both numerical and categorical data.
No need for data normalization.
Fast to train and predict.
Can show which features are most important.
5. Limitations
Can overfit if the tree becomes too deep.
Small changes in data may create a different tree.
Less accurate than ensemble methods like Random Forest for complex datasets.
May become complex for very large datasets.
6. Python Example
Python
Run
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
Optimize Prompt
DecisionTreeClassifier
Ask anything

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ChatGPT
Get Plus
Share
New chat
Library
Projects
Plugins
Codex
More
Pinned
RPA Development Help
MediScript AI System
Recents
Decision Tree with Iris Dataset
Prompt Engineering Showdown
Message Categorization Task
AI/ML Internship Application
Resume Optimization for Web D...
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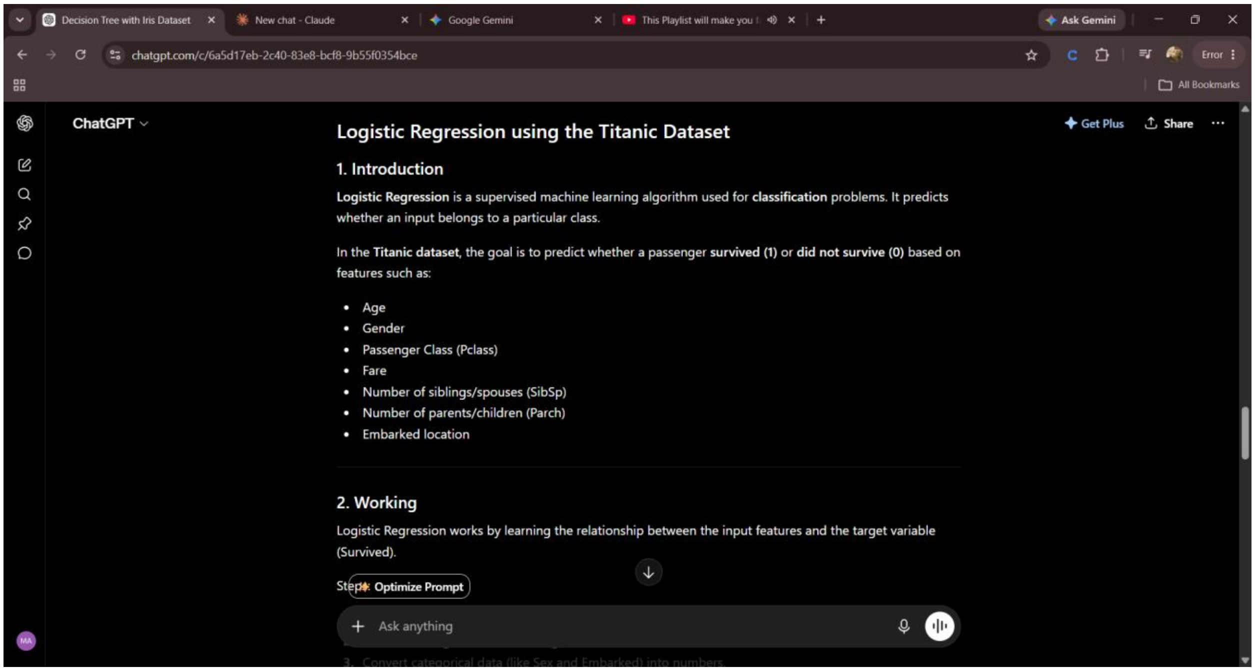
Free
Upgrade
7. Applications
Decision Trees are widely used in:
Medical diagnosis
Customer segmentation
Loan approval systems
Fraud detection
Student performance prediction
Weather forecasting
Disease classification
Plant and flower classification
8. Interview Questions
1. What is a Decision Tree?
2. Is Decision Tree supervised or unsupervised learning?
3. What is the difference between Gini Index and Entropy?
4. Why can Decision Trees overfit?
5. How can overfitting be reduced in a Decision Tree?
6. Does Decision Tree require feature scaling?
7. What are leaf nodes and decision nodes?
8. Why is the Iris dataset suitable for Decision Trees?
Optimize Prompt
Ask anything

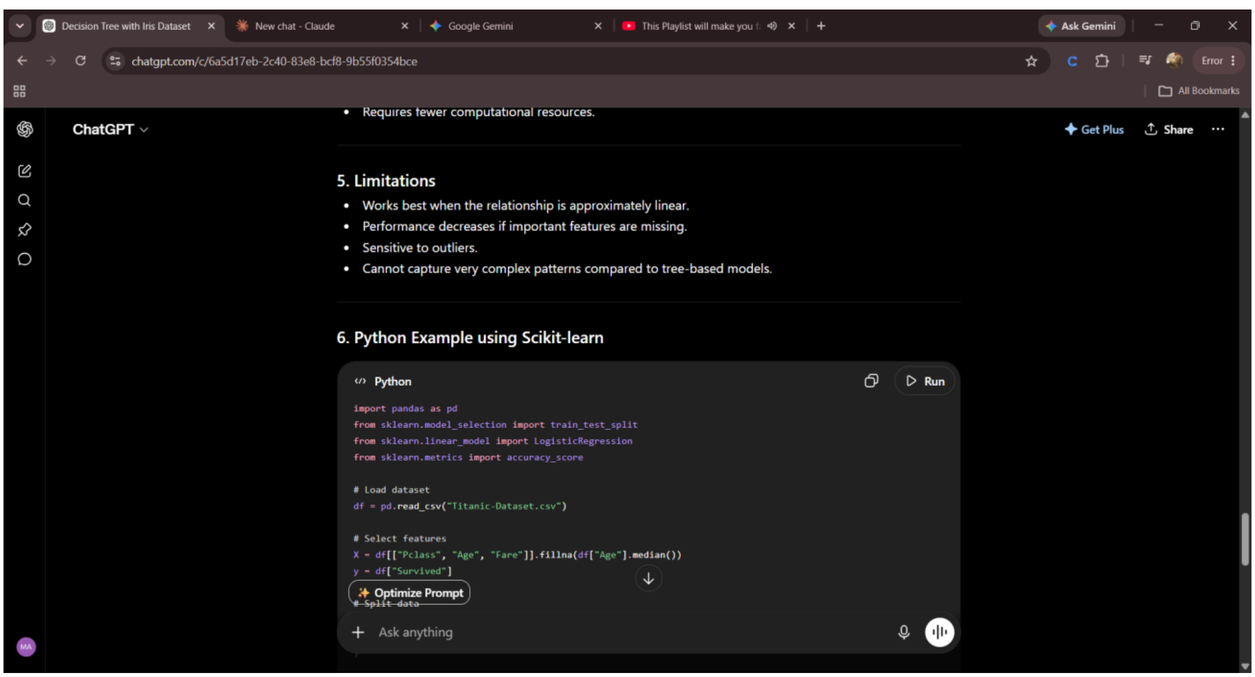
Prompt 2

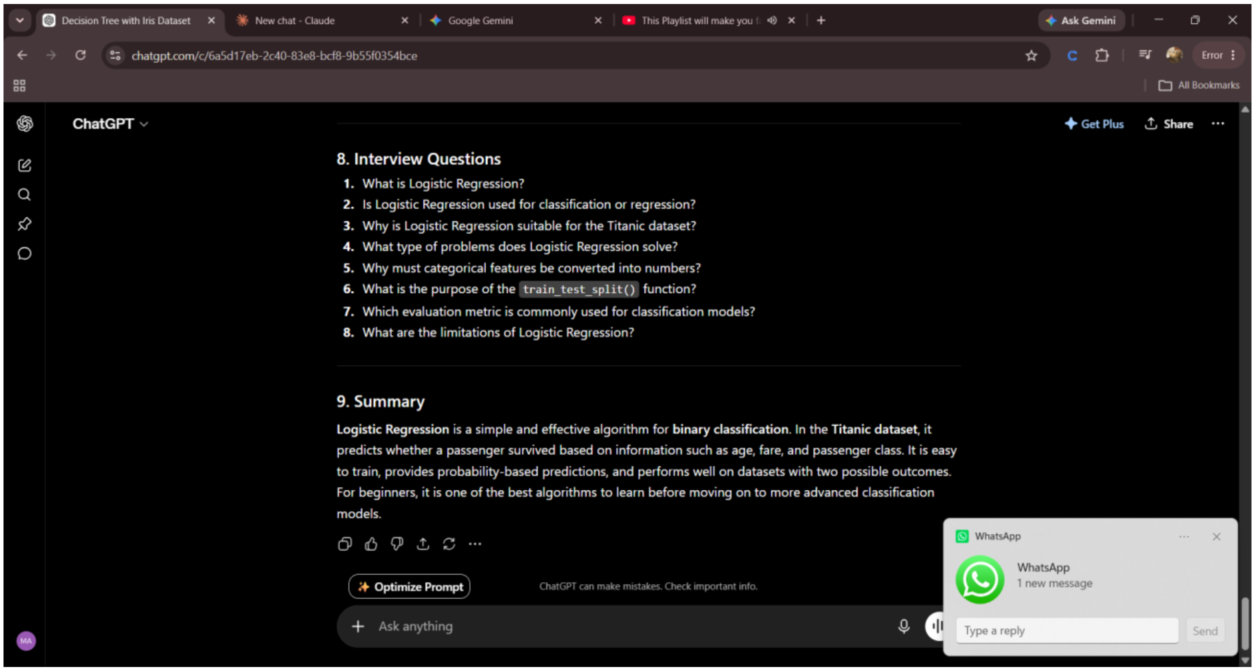
Variables

Dataset: Titanic Dataset

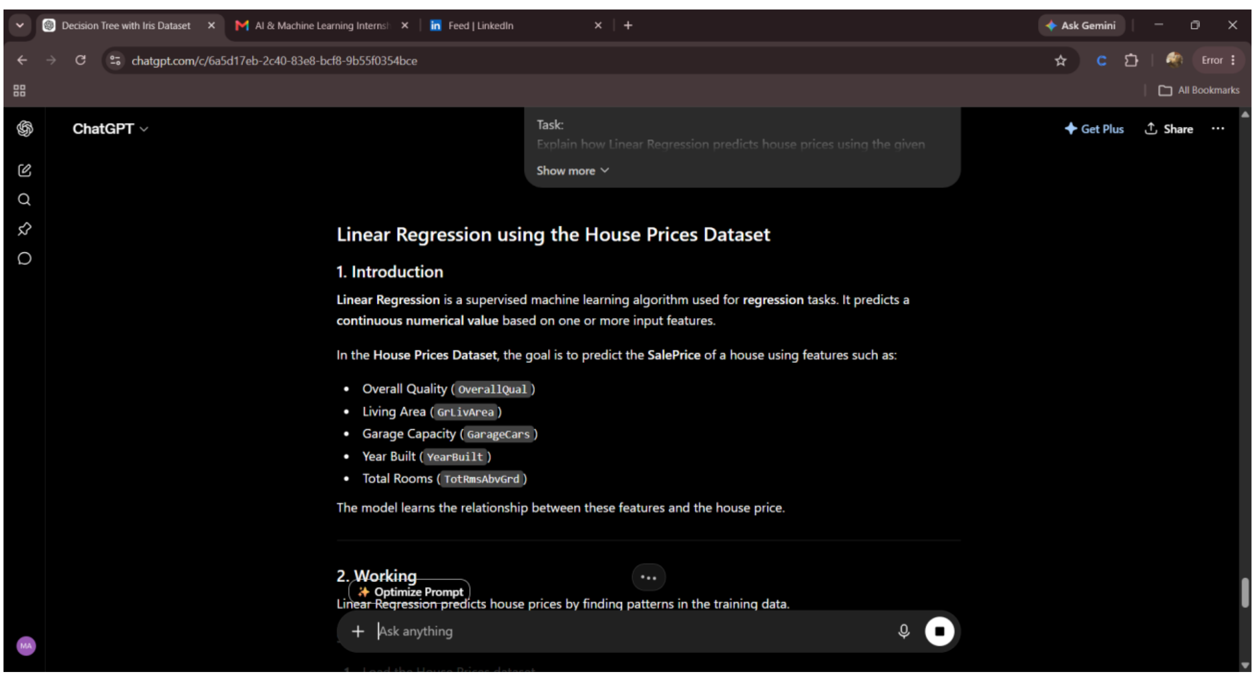
Algorithm: Logistic Regression

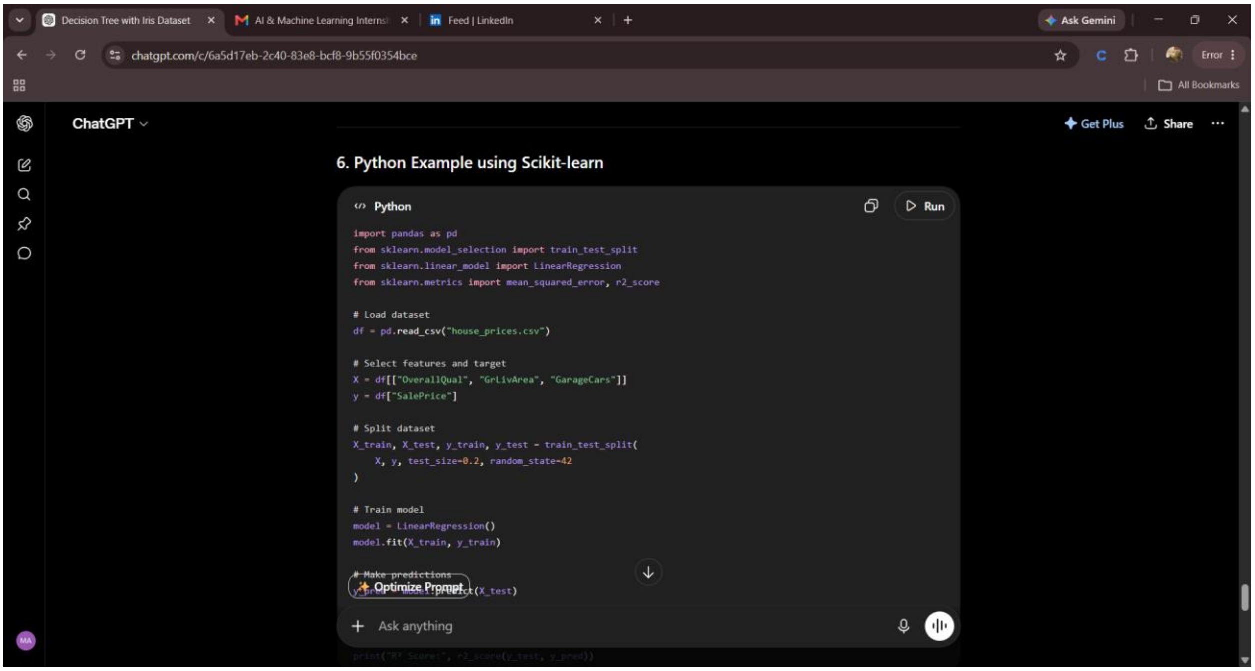


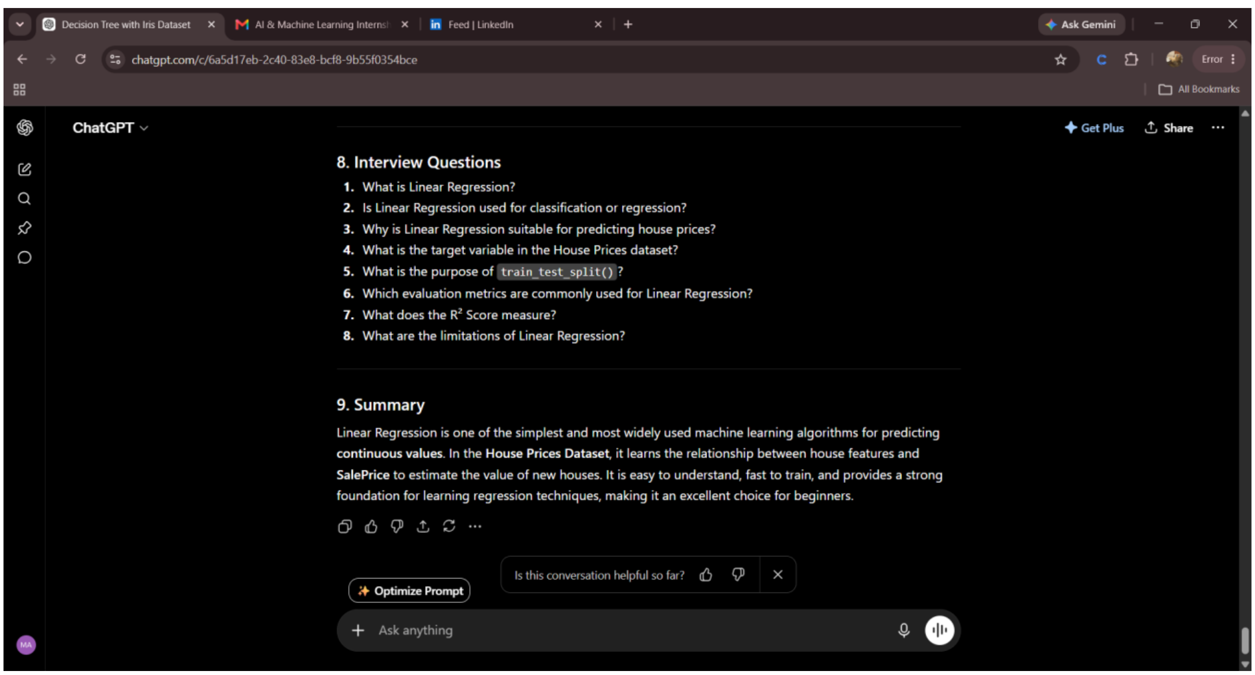




Prompt 3 – House Prices + Linear Regression







Prompt 4 – Mall Customers + K-Means Clustering

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K-Means Clustering using the Mall Customers Dataset

1. Introduction

K-Means Clustering is an **unsupervised machine learning** algorithm used to group similar data points into clusters. Unlike supervised learning, it does **not require labeled data**.

In the **Mall Customers Dataset**, the goal is to group customers based on their shopping behavior using features such as:

- Annual Income
- Spending Score

The algorithm helps businesses identify different types of customers for better marketing and decision-making.

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2. Working

K-Means Clustering groups customers based on the similarity of their features.

Steps:

1. Load the Mall Customers dataset.
2. Select important features (such as Annual Income and Spending Score).

Optimize Prompt

Ask anything

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Python | Run

```python
kmeans = KMeans(n_clusters=5, random_state=42)
df["Cluster"] = kmeans.fit_predict(X)

# Plot clusters
plt.scatter(X["Annual Income (k$)"], X["Spending Score (1-100)"],
            c=df["Cluster"], cmap="viridis")
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score")
plt.title("Customer Segments")
plt.show()
```

---

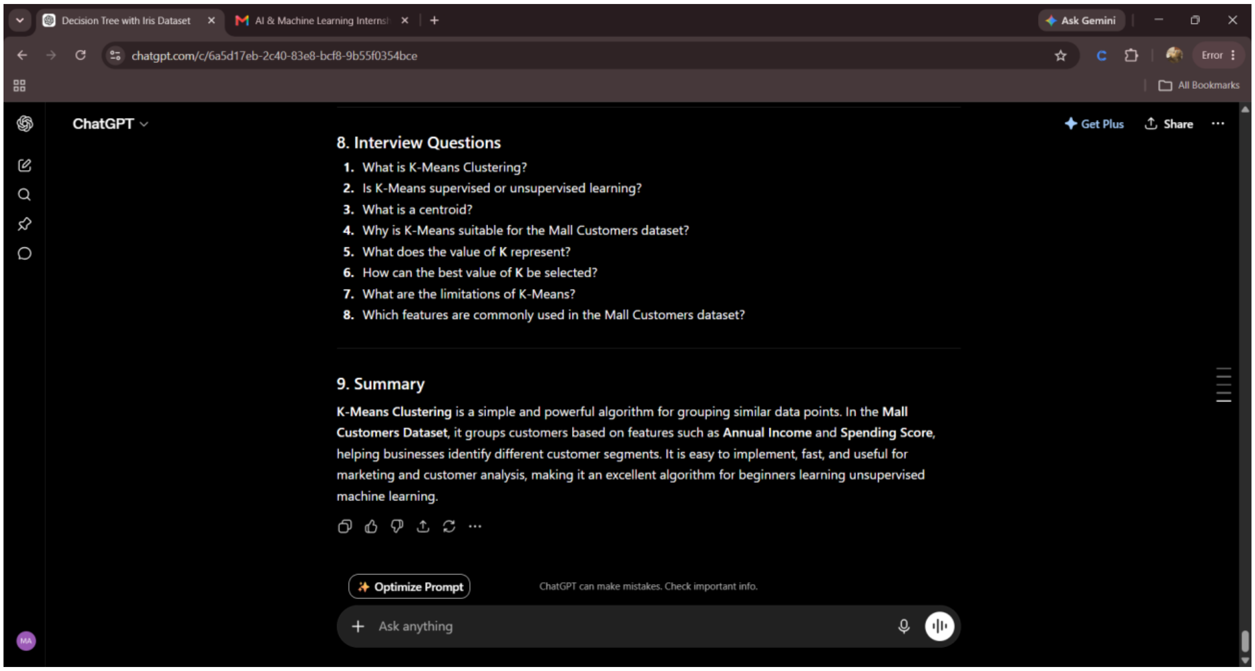
7. Applications

K-Means Clustering is widely used in:

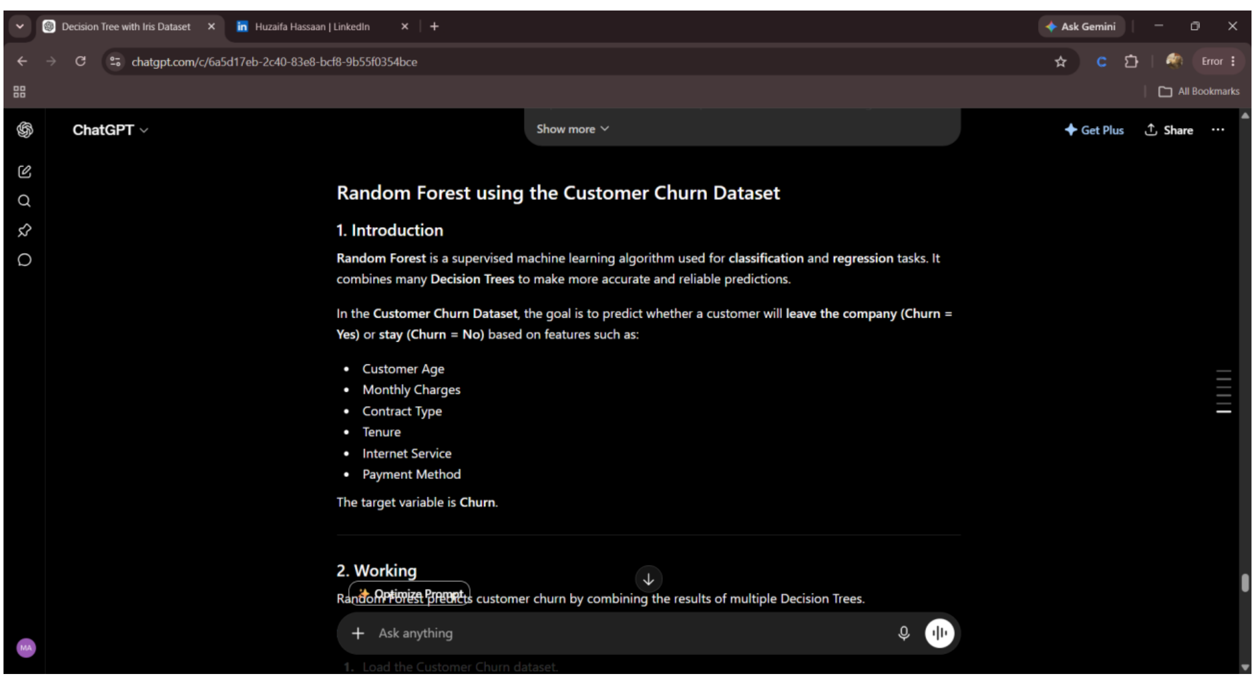
- Customer segmentation
- Market analysis
- Product recommendation
- Sales and marketing strategies
- Image segmentation
- Fraud detection
- Social media user analysis

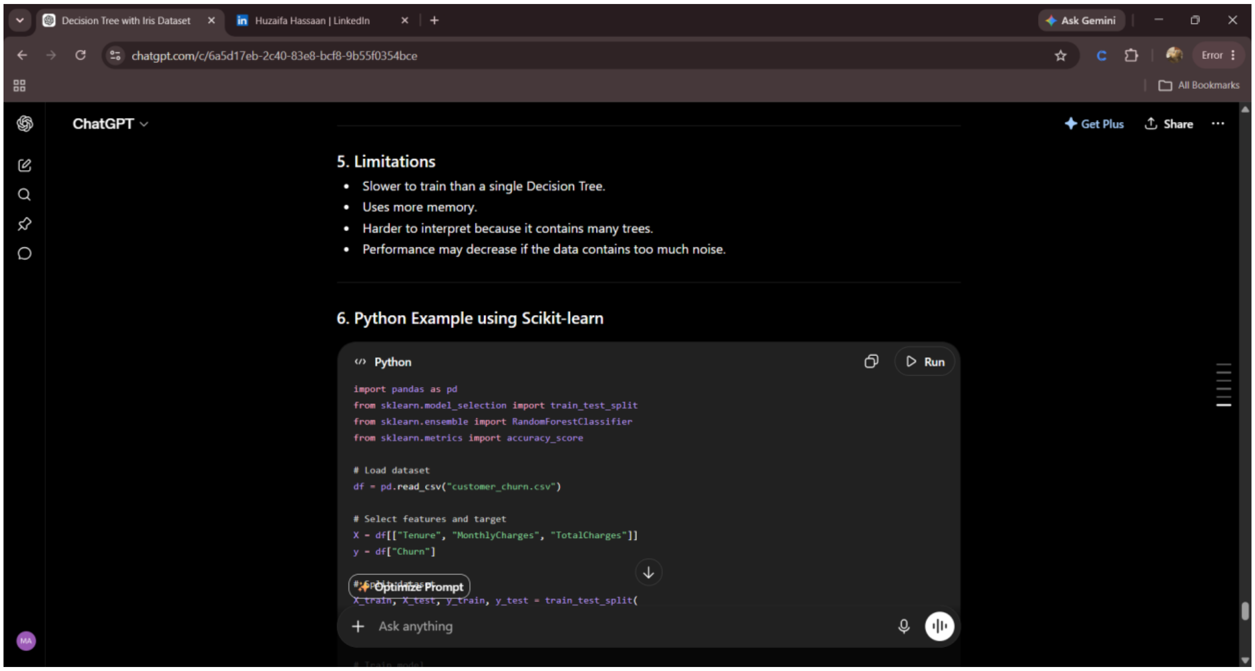
Optimize Prompt

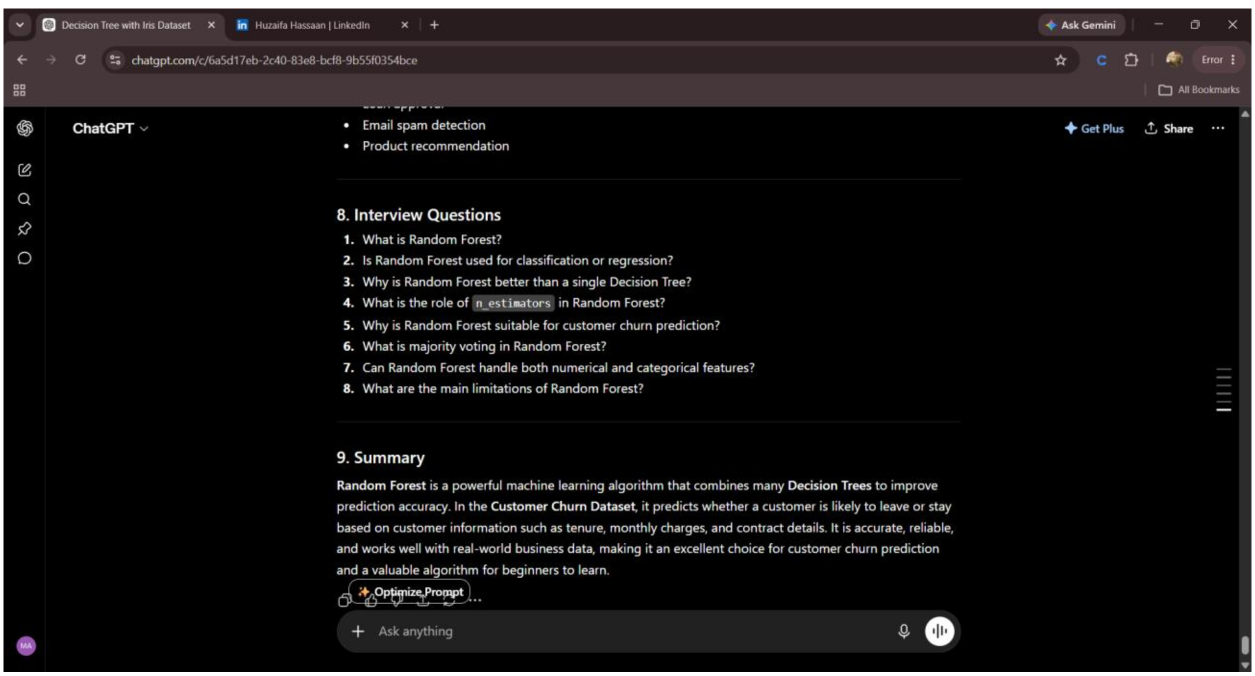
Ask anything



Prompt 5 – Customer Churn + Random Forest







Key Learning

This prompt library demonstrates how one reusable template can be adapted for multiple AI and Machine Learning topics by changing only the dataset and algorithm. Using a structured prompt improves consistency, saves time, and produces well-organized responses. Reusable prompt templates are an effective way to standardize interactions with AI models and support efficient learning.

Conclusion

Creating reusable prompt templates is a fundamental prompt engineering technique. Rather than writing prompts from scratch, developers can design structured templates with placeholders and reuse them across different scenarios. This improves productivity, consistency, and the overall quality of AI-generated responses.